

European service-hub expansion

Board decision brief | synthetic client data | frozen official-series archive

Recommendation

Approve a staged ceiling of €225,000 year-zero capex and five FTE for Spain and Czechia. Commit Spain first (€130,000; three FTE); reserve Czechia (€95,000; two FTE) until the day-60 operating and commercial gate. The pair has €198,099 modeled annual incremental contribution after recurring fixed costs in the client base case and a 1.14-year simple payback. The defined severe joint stress reduces its annual gain to €32,180 and extends simple payback to 6.99 years. [C1, C2, A2]

Decision frame and status

Question	Hard conditions	Decision basis
Fund up to two of six local hubs, or defer.	≤€450k capex; ≤7 FTE; ≤2 hubs.	Primary planning criterion: incremental annual contribution after recurring fixed cost; then resilience, cash exposure and staged execution. The board has not supplied numerical weights.
Time boundary	2025 shipped cohort, returns known through 31 Jan 2026.	Hub estimates are annual run-rate assumptions, not observed causal effects or a discounted cash-flow forecast.

Conclusion card. The base-case choice is CZE+ESP; the strongest stress alternative is NLD+ESP. The choice accepts Czechia's PLN/CZK-style FX sensitivity and unproven 25% volume uplift. If roughly 29% of the defined joint stress materializes on the second-hub comparison, NLD replaces CZE on annual incremental contribution; a measured Czechia pilot below the NLD-equivalent economics also reopens the choice. The actual uplift, savings and operating costs need validation before Czechia capex is released. [C2, A2]

1 Evidence-linked diagnosis

The resolved 2025 cohort contains 1,254 orders and 77,436 units. It booked €4,464,622 gross sales and €80,831 credits, yielding €4,383,792 net sales. After €2,246,919 net COGS and €106,862 nonrefundable fulfillment, contribution is €2,030,010. All 72 country-month records reconcile to the six totals below. [C1, A1]

Market	Orders	Units	Net sales €	Contribution €	Margin	Returned units
DEU	209	12,153	625,792	262,366	41.9%	73
FRA	209	11,880	632,231	276,271	43.7%	144
NLD	209	12,576	697,733	315,644	45.2%	144
POL	209	13,163	761,890	356,494	46.8%	180
CZE	209	13,733	815,307	397,335	48.7%	288
ESP	209	13,931	850,839	421,900	49.6%	252
TOTAL	1,254	77,436	4,383,792	2,030,010	46.3%	1,081

Spain and Czechia have the highest 2025 contribution among these synthetic country cohorts; each market happens to have 209 eligible orders, so the differences arise from units, price/cost and returns rather than observed order counts. This is a client-specific financial signal, not a population-based demand estimate. [C1]

Booked net sales equal shipped goods value less eligible credits, translated at the original sale-month reference quote. Cash receipts and payments may occur on different dates and at actual bank rates; no settlement ledger or receivable balance was supplied. Contribution further deducts product cost net of only restocked units and actual fulfillment. It omits corporate overhead, taxes, hub startup capex and recurring hub fixed cost, which enter the option model separately. [A1]

Data reconciliation and quality

Across two pages and corrections there are 1,328 raw order rows, 1,297 distinct IDs, 13 exact repeats and 18 superseded revision rows. Highest numeric revision wins; 24 cancelled, 18 test and one January 2026 shipment are excluded. The 24 raw missing shipment dates belong to cancelled rows. All 12 valid free shipments stay in units, COGS and fulfillment. [C1, A1]

Returns have 264 raw rows, 243 IDs, 20 exact repeats and one superseded revision. 241 resolved return IDs enter the 2025 shipped cohort; one orphan is quarantined and one 1 Feb 2026 return is after the inclusive cutoff. The included 1,081 returned units yield only 481 reusable units for COGS recovery. The order/return audit and ledger preserve every decision. [C1, A1]

2 Market context and exchange rates

These are contextual public observations, not evidence of assembly demand or a causal hub uplift. The World Bank archive was collected by the source room on 27 Sep 2026 and labels the series last updated 13 Jul 2026. Values may contain later revisions of 2022–2024 observations. [P1, P2]

Market	Population 2022	Population 2024	Change	2024 GDP/capita, current US\$
DEU	83,177,813	83,516,593	0.41%	56,104
FRA	68,184,457	68,551,653	0.54%	46,103
NLD	17,700,982	17,993,485	1.65%	67,465
POL	36,821,749	36,559,233	-0.71%	25,104
CZE	10,672,118	10,905,028	2.18%	31,828
ESP	47,786,102	48,848,840	2.22%	35,327

Spain has the largest 2022–2024 population increase in this set (+2.22%); Poland declined (−0.71%). The Netherlands' 2024 current-US\$ GDP per capita is highest here, while Poland's is lowest. These series help frame size and economic context but do not measure Meridian's installed base, service failures, willingness to pay or local delivery advantage. All 18 country-year combinations are present in the archive; the workbook retains each year and underlying values. [P1, P2]

The 2025 ECB archive supplies all 12 monthly means for PLN and CZK. The quoted direction is local currency units per EUR; the model divides local gross sales and aggregate local refunds by the original sale month's mean. The 2025 PLN monthly means range from 4.1722 to 4.2658; CZK from 24.2340 to 25.1633. EUR orders use 1. These means are analytical translations, not actual transaction settlement rates. The workbook lists all 24 means and observation counts. [E1, A1]

3 Quantified option screen

Client-owned synthetic assumptions: volume uplifts 10%/25%/40%; per-unit savings and recurring fixed costs from the option file. Annual incremental contribution = baseline contribution × uplift + baseline units × (1+uplift) × savings per unit − recurring fixed cost. Year-zero capex is separate; simple payback divides capex by positive annual gain. The joint stress adds a refund shock of 3% of gross sales without further cost recovery, plus a 10% net-sales depreciation shock in PLN/CZK markets. [C2, A2]

Hub	Capex €	FTE	Low €/yr	Base €/yr	High €/yr	Stress €/yr
DEU	280,000	5	-34,006	10,089	54,183	-13,583
FRA	210,000	4	-19,476	27,133	73,741	3,098
NLD	140,000	3	10,832	64,215	117,598	37,623
POL	110,000	3	25,400	83,612	141,825	-40,743

Hub	Capex €	FTE	Low €/yr	Base €/yr	High €/yr	Stress €/yr
CZE	95,000	2	31,457	95,383	159,309	-37,827
ESP	130,000	3	33,162	102,716	172,270	70,007

All six singles meet the hard budget and staffing limits. Of 15 possible pairs, 11 are feasible; each pair adds its constituent country economics with no synergy. The workbook includes every single, all pairs, defer, all four scenarios, feasibility and payback. Negative annual contribution has no payback. [C2, A2]

Feasible pair	Base €/yr	Stress €/yr	Capex €	FTE	Base payback y
CZE+ESP	198,099	32,180	225,000	5	1.14
POL+ESP	186,329	29,264	240,000	6	1.29
POL+CZE	178,995	-78,570	205,000	5	1.15
NLD+ESP	166,931	107,630	270,000	6	1.62
NLD+CZE	159,598	-203	235,000	5	1.47
NLD+POL	147,828	-3,119	250,000	6	1.69
FRA+ESP	129,849	73,105	340,000	7	2.62
FRA+CZE	122,516	-34,729	305,000	6	2.49
FRA+POL	110,745	-37,645	320,000	7	2.89
DEU+CZE	105,472	-51,410	375,000	7	3.56
FRA+NLD	91,348	40,721	350,000	7	3.83

4 Choice, alternatives and switching test

The hard constraints are capex, staffing and hub count. Among feasible choices, CZE+ESP is first on annual gain in low (€64,619), base (€198,099) and high (€331,580). It consumes €225,000, five FTE and two hubs, leaving €225,000 budget headroom and two FTE. Its base gain is €11,770 above the second-ranked base pair POL+ESP; the stress gain is €2,916 higher than POL+ESP but still thin relative to capex. [C2]

The more consequential alternative is NLD+ESP. It gives up €31,168 annual base gain and needs €45,000 more capex and one more FTE, but delivers €107,630 under the defined joint stress versus €32,180 for CZE+ESP. It avoids the modeled CZK depreciation shock. In the stress criterion, NLD+ESP dominates the selected pair; in base annual gain and capital efficiency, CZE+ESP is stronger. Neither dominates across all criteria. Deferring has zero capex and zero modeled incremental gain, and remains sensible if uplift or savings fail validation. [C2]

The base-vs-stress second-hub edge changes from +€31,168 for Czechia to -€75,450 under the full defined joint stress. Linear interpolation of that specific stress bundle crosses zero at about 29.2% of the bundle. This is a scenario sensitivity, not a probability, confidence interval or forecast. It provides the board a measurable switch trigger: if updated, evidenced Czechia refund/FX and unit economics imply a worse annual second-hub gain than the Netherlands on the same basis, select NLD or defer the second hub. [C2, A2]

5 First 90 days: staged execution

Days	Accountable owner	Dependency and work	Decision gate / evidence
0–15	COO + Finance	Approve Spain spend ceiling; validate 2025 ledger, carrier/service baseline and site quote. Ring-fence Czechia spend only.	Signed costed Spain charter; monthly country ledger reconciles to source; legal and tax checks assigned.
16–30	Spain operations lead	Finalize lease/3PL and inventory layout; define SKU stocking and reverse logistics with Finance.	Contracted annual fixed cost and savings per shipped unit remain within model tolerance; launch readiness review.
31–60	Spain lead + Commercial	Pilot Spain routing and scan events; run Czechia carrier and customer-order feasibility sample from real operations.	≥95% on-time dispatch, ≥98% scan completeness; verified unit savings and return handling. CFO compares refreshed CZE vs NLD annual gain.
61–90	COO + CFO + board sponsor	Scale Spain if service and economics hold; release CZE only after gate; otherwise compare NLD or defer.	CZE release requires a positive risk-adjusted case, documented FX/return sensitivity, signed fixed-cost quote and capacity plan; board records decision.

KPI plan

Metric	Definition / cadence	Owner and action threshold
Delivery	On-time dispatch ÷ eligible hub shipments, weekly; compare to central baseline for same destinations/SKUs.	Spain lead; ≥95% pilot target, investigate any 2-week shortfall.
Unit economics	Actual carrier + handling EUR per shipped unit against pre-launch country baseline, weekly.	Finance; verify at least the option-file €3.0 Spain and €2.1 Czechia unit saving before scale.
Returns	Returned units ÷ shipped units by sale cohort; refund EUR ÷ gross sales EUR, monthly.	Finance + Quality; re-run 3% gross-sales shock and quarantine unmatched returns.
Contribution	Order-rounded net sales – net COGS – fulfillment – annualized hub fixed cost, monthly.	CFO; positive forward annualized gain and refreshed payback before second commitment.
Data quality	Unmatched return IDs, duplicate IDs, missing cost/FX, monthly-to-country difference.	Data lead; unresolved unmatched records remain quarantined; monetary reconciliation target €0.

Owners are proposed roles, not named client personnel. KPI thresholds are implementation controls, not observations from the source room. A 90-day pilot cannot prove a full-year uplift; continuation requires updated run-rate evidence and a conservative capacity check. [A2]

6 Risks, evidence status and limitations

Risk	Why it matters	Control / owner
Volume uplift is assumed	The 10/25/40% cases are client planning inputs, with no observed hub experiment or demand elasticity.	Commercial and Finance compare routed cohorts with the central baseline; do not interpret modeled gain as causal.
FX and refund stress	Czechia's modeled annual gain turns negative under the combined shock; actual settlement FX is unavailable.	CFO refreshes monthly FX and refund sensitivity before CZE release; NLD+ESP is the stress fallback.
Fixed-cost/site quote risk	A higher lease/3PL fixed cost lowers annual gain euro-for-euro; capex overspend consumes headroom.	COO obtains binding quotes and logs committed versus authorized capex.
Service and stock-outs	A hub can raise inventory burden or miss service promises despite forecast savings.	Spain lead pilots SKU assortment, scan completeness and dispatch service before scale.
Data scope	Returns after 31 Jan 2026 and transaction cash timing are outside the cohort; orphan return is not attributed.	Data lead refreshes cohort roll-forward and keeps unmatched records separate.

Evidence register (saved locally in evidence/raw and evidence/source_register.csv)

[C1] Synthetic client exports: orders-part1.csv, orders-part2.csv, order-corrections.csv, returns.csv, unit-costs.csv; full hashes, collection timestamps and units in source register.

[C2] Synthetic client hub-options.csv; capex, FTE, fixed cost and saving EUR/unit.

[A1] Synthetic client data-dictionary.md; grain, revision, currency, cost and return rules.

[A2] Synthetic client scenario-policy.md; uplift, stress, pair and payback rules.

[P1] Archived World Bank population, SP.POP.TOTL, 2022–2024:

https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000

[P2] Archived World Bank GDP per capita current US\$, NY.GDP.PCAP.CD, 2022–2024:

https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000

[E1] Archived ECB euro reference history ZIP: <https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip>; extracted CSV saved in source room.

Archive versus live research.

The analysis copied the frozen source room at <http://127.0.0.1:65178/> and verified public archive file hashes against its provenance metadata. It did not query live ECB or World Bank endpoints for the core calculations. The copied public responses were archived on 27 Sep 2026; local copy times, hashes, periods and units appear in the source register. No customer interviews, actual settlement rates, demand study, route-time baseline, site quotes or operational pilot outcomes were supplied. The board recommendation is conditional on those missing operating inputs, rather than statistically proven.

[C1, C2, P1, P2, E1]